

Pythagoras and Trigonometry

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Complete Pythagoras' theorem for a right-angled triangle with shorter sides a and b and hypotenuse c .

$$a^2 + \square = \square$$

.....

2 [2 marks]

A right-angled triangle has shorter sides 6 cm and 8 cm.

Work out the length of the hypotenuse.

.....

3 [2 marks]

A right-angled triangle has shorter sides 9 cm and 12 cm.

Work out the length of the hypotenuse.

.....

4 [2 marks]

A right-angled triangle has hypotenuse 13 cm and one shorter side 5 cm.

Work out the length of the other shorter side.

.....

5 [3 marks]

A right-angled triangle has shorter sides 7 cm and 10 cm.

Work out the hypotenuse, correct to 1 decimal place.

.....

6

[3 marks]

A right-angled triangle has hypotenuse 15 cm and one shorter side 9 cm.

- (a) Work out the other shorter side.
 - (b) Work out the area of the triangle.
-

7

[3 marks]

In a right-angled triangle the hypotenuse is 10 cm and one angle is 40° .

Work out the side opposite the 40° angle, to 1 decimal place.

8

[3 marks]

In a right-angled triangle the hypotenuse is 12 cm and one angle is 25° .

Work out the side adjacent to the 25° angle, to 1 decimal place.

9

[4 marks]

In a right-angled triangle, the side opposite angle θ is 5 cm and the hypotenuse is 9 cm.

Work out the size of angle θ , to 1 decimal place.

10

[4 marks]

A ladder of length 5 m leans against a vertical wall. The foot of the ladder is 1.4 m from the base of the wall.

- (a) How far up the wall does the ladder reach? Give your answer to 1 decimal place.
 - (b) Work out the angle between the ladder and the ground, to 1 decimal place.
-

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1 [2 marks]

A right-angled triangle has hypotenuse 17 cm and one shorter side 8 cm.
Work out the other shorter side.

.....

2 [2 marks]

Work out the distance between the points (1, 2) and (5, 5).

.....

3 [3 marks]

In a right-angled triangle the side opposite angle θ is 8 cm and the side adjacent is 6 cm.
Work out θ , to 1 decimal place.

.....

4 [3 marks]

Write down the exact value of
(a) $\sin 30^\circ$ (b) $\cos 60^\circ$ (c) $\tan 45^\circ$

.....

5 [4 marks]

A cuboid measures 6 cm by 8 cm by 10 cm.
Work out the length of the longest straight line that can be drawn inside it, to 1 decimal place.

.....

6 [4 marks]

In triangle ABC, angle $A = 50^\circ$, angle $B = 35^\circ$ and side a (opposite A) = 12 cm.
Work out side b , to 3 significant figures.

.....

7

[4 marks]

A triangle has sides 5 cm, 7 cm and 9 cm.

Work out the size of the largest angle, to 1 decimal place.

8

[4 marks]

A triangle has two sides of 8 cm and 11 cm with an angle of 47° between them.

Work out the area of the triangle, to 3 significant figures.

9

[4 marks]

A ship sails 40 km on a bearing of 070° , then 30 km on a bearing of 160° .

(a) Work out how far the ship now is from its starting point.

(b) Work out the bearing of the ship from its starting point, to the nearest degree.

10

[5 marks]

A pyramid has a square base ABCD of side 8 cm. The apex E is directly above the centre of the base, and $EA = 10$ cm.

(a) Work out the length of the base diagonal AC, to 2 decimal places.

(b) Work out the vertical height of the pyramid, to 2 decimal places.

(c) Work out the angle EA makes with the base, to 1 decimal place.
